Quantum Computing and Option Pricing

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2025-2026

Outline

The origins of quantum computing

Richard Feynman (1981)

"*Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical*"

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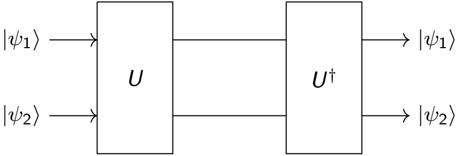
Some milestones

Benioff, 1980	Quantum mechanical model of a computer
Bennet, Brassard, 1984	First quantum cryptography protocol
Deutsch, 1985	Quantum Turing Machine
Shor, 1994	Prime factorization

Reversibility of quantum operations

All operations on qubits are reversible



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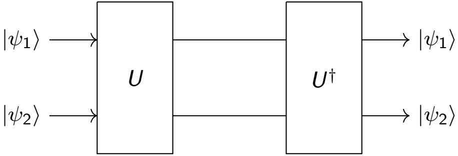
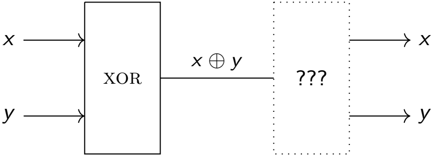


But most logical operations are not



How can algorithms be conceived if most logical operations are unavailable?

The no-cloning theorem

Is it possible to copy a qubit? In other words, does there exist a unitary matrix U such that for all $|\psi\rangle$, $U \cdot |\psi\rangle = |\psi\rangle \otimes |\psi\rangle$?

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Theorem

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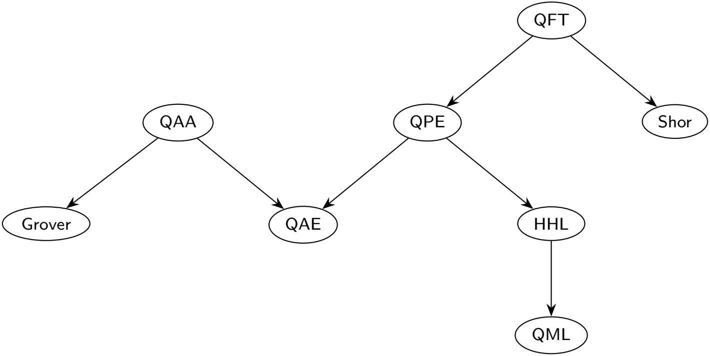
Proof Assume for the sake of contradiction that such an operation U exists, and let $|\psi_1\rangle$ and $|\psi_2\rangle$ be arbitrary qubits:

$$\begin{aligned}
\langle\psi_1|\psi_2\rangle &= \langle\psi_1|\psi_2\rangle \cdot \langle 0|0\rangle \\
&= (\langle\psi_1| \otimes \langle 0|) \cdot (|\psi_2\rangle \otimes |0\rangle) \\
&= (|\psi_1\rangle \otimes |0\rangle)^\dagger \cdot U^\dagger \cdot U \cdot (|\psi_2\rangle \otimes |0\rangle) \\
&= (U \cdot (|\psi_1\rangle \otimes |0\rangle))^\dagger \cdot (U \cdot (|\psi_2\rangle \otimes |0\rangle)) \\
&= (|\psi_1\rangle \otimes |\psi_1\rangle)^\dagger \cdot (|\psi_2\rangle \otimes |\psi_2\rangle) \\
&= \langle\psi_1|\psi_2\rangle \cdot \langle\psi_1|\psi_2\rangle
\end{aligned}$$

Under the assumption that U exists, for all $|\psi_1\rangle$ and $|\psi_2\rangle$ we have $\langle\psi_1|\psi_2\rangle = \langle\psi_1|\psi_2\rangle^2$, a contradiction

A large collection of quantum algorithms

- Quantum Fourier Transform
- Quantum Phase Estimation
- Quantum Amplitude Amplification
- Quantum Amplitude Estimation
- Grover's algorithm
- Shor's algorithm
- Harrow-Hassidim-Lloyd (HHL) Algorithm
- Quantum Machine Learning



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- **Quantum circuits**
  - A generalization of boolean circuits
  - Nb: the horizontal axis of a circuit represents time, not space

Quantum circuits

A quantum circuit consists of

- Lines (or wires), representing qubits
- Double lines, representing classical data
- Gates, representing unitary operations
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Examples

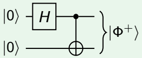

Figure: Bell state generation

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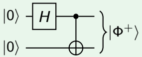


Figure: Bell state generation

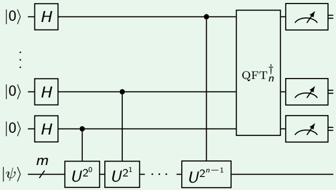
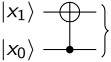


Figure: Quantum Phase Estimation

Some gates involving two qubits

CNOT



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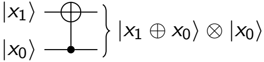
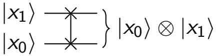


SWAP



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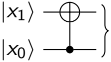

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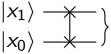

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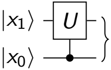
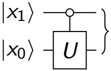
Controlled operation



$|x_1\rangle \otimes |x_0\rangle$ if $x_0 = 0$ and $(U \cdot |x_1\rangle) \otimes |x_0\rangle$ if $x_0 = 1$



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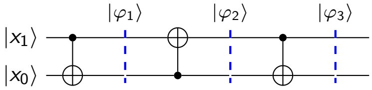
Consider the operation U represented by the circuit below. What elementary operation does U represent?



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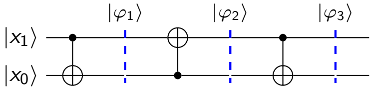

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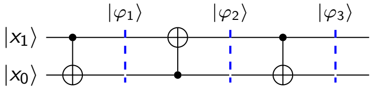

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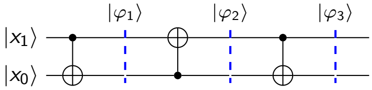

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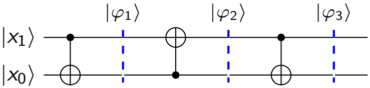

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U represents the SWAP gate

Quantum teleportation

- About quantum teleportation
  - Transfer of quantum information using classical information
  - The “opposite” of superdense coding
  - Initially proposed by Bennett, Brassard et al. (1993)
  - Experimentally verified on up to 1400 km

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- About quantum teleportation
  - Transfer of quantum information using classical information
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  - Experimentally verified on up to 1400 km
- Principle of the protocol
  - Alice wishes to transmit a qubit state $|\psi\rangle_{A_1} = \alpha|0\rangle_{A_1} + \beta|1\rangle_{A_1}$ to Bob using two classical bits
  - Alice and Bob share a Bell state $|\Phi^+\rangle_{A_2B} = \frac{1}{\sqrt{2}}\left(|0\rangle_{A_2}|0\rangle_B + |1\rangle_{A_2}|1\rangle_B\right)$
  - Alice can apply any transformation to her qubits before sending the two bits of information to Bob
  - Upon receiving the two bits of information, Bob can apply any transformation to his part of the Bell state

Quantum teleportation protocol

- On Alice's side
  - She applies CNOT to her qubits
  - She applies H to the first qubit
  - She measures her qubits and sends the classical output a_0, a_1 to Bob

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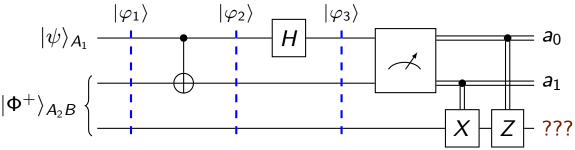

Analysis of Alice's operations

- Initial state

$$|\varphi_1\rangle = |\psi\rangle_{A_1} \otimes |\Phi^+\rangle_{A_2 B} = (\alpha|0\rangle_{A_1} + \beta|1\rangle_{A_1}) \otimes \left(\frac{1}{\sqrt{2}}\left(|0\rangle_{A_2}|0\rangle_B + |1\rangle_{A_2}|1\rangle_B\right)\right)$$

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- Application of $\text{CNOT}_{A_1A_2} \otimes \mathbf{I}_B$ to $|\varphi_1\rangle$

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&= \frac{1}{\sqrt{2}}\left(\alpha|0\rangle|0\rangle|0\rangle + \beta|1\rangle|1\rangle|0\rangle + \alpha|0\rangle|1\rangle|1\rangle + \beta|1\rangle|0\rangle|1\rangle\right) \\
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- Measurement: Alice reads and sends the output to Bob, whose qubit is in state $|\psi_4\rangle$

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The Deutsch and Deutsch-Jozsa problems

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- Both algorithms rely on the following fact:

Property

Given $f : \mathbb{Z}_2^n \to \mathbb{Z}_2$ and $\vec{x} \in \mathbb{Z}_2^n$, we have $|f(\vec{x})\rangle - |1 \oplus f(\vec{x})\rangle = (-1)^{f(\vec{x})}(|0\rangle - |1\rangle)$

Unitary oracles

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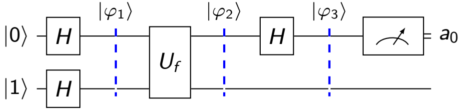
- In other words, a single application of U_f can generate a superposition of **all** the images of elements in $\mathbb{Z}_2^n$ by f
- But only a single output can be nondeterministically read; it is necessary to carry out additional operations to extract relevant information

Deutsch's algoritm



$$|\varphi_1\rangle = \frac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle - |1\rangle) = \frac{1}{2}\left(|0\rangle \otimes (|0\rangle - |1\rangle) + |1\rangle \otimes (|0\rangle - |1\rangle)\right)$$

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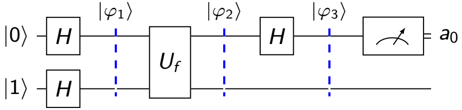

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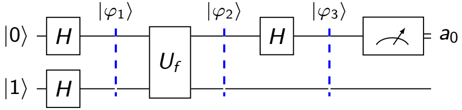

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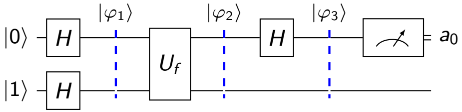
Two cases:

Deutsch's algoritm



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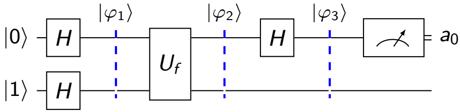
Deutsch's algoritm



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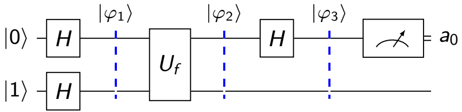
Deutsch's algoritm (cont'd)



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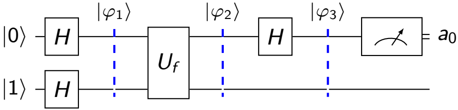
  - The measurement of the first qubit outputs "0" with probability $|k|^2 = 1$

Deutsch's algoritm (cont'd)



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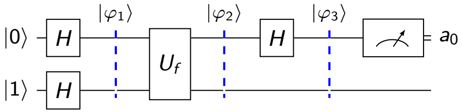

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Conclusion

The single query to U_f permits to determine whether f is constant or balanced

The Deutsch-Jozsa algorithm: preliminary results

Proposition

For all $n \geq 1$ and for all $\vec{x} \in \mathbb{Z}_2^n$, we have

$$H^{\otimes n} \cdot |\vec{x}\rangle = \frac{1}{\sqrt{2^n}} \sum_{\vec{y} \in \mathbb{Z}_2^n} (-1)^{\vec{x} \odot \vec{y}} |\vec{y}\rangle, \text{ were } \vec{x} \odot \vec{y} = \sum_{i=0}^{n-1} x_i y_i$$

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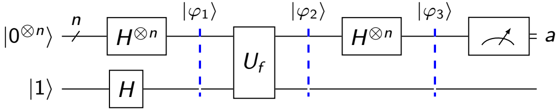
Remark

By identifying the binary notation $y_{n-1} \cdots y_0$ with the number $y = \sum_{i=0}^{n-1} 2^i y_i$, it is common to write

$$H^{\otimes n} \cdot |0^n\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} |y\rangle$$

The Deutsch-Jozsa algorithm



$$\varphi_1 = (H^{\otimes n} \otimes H) \cdot (|0^n\rangle \otimes |1\rangle) = \frac{1}{\sqrt{2^{n+1}}} \left(\sum_{\vec{x} \in \mathbb{Z}_2^n} |\vec{x}\rangle \right) \otimes (|0\rangle - |1\rangle)$$

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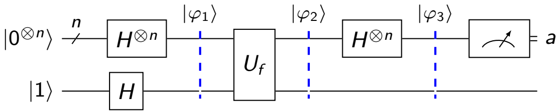
The Deutsch-Jozsa algorithm



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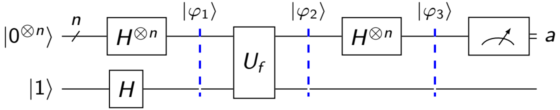
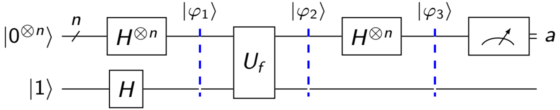
The Deutsch-Jozsa algorithm (cont'd)



$$\varphi_3 = \frac{1}{\sqrt{2^{n+1}}}\left(\sum_{\vec{x}\in\mathbb{Z}_2^n}(-1)^{f(\vec{x})} H^{\otimes n}\cdot|\vec{x}\rangle\right)\otimes(|0\rangle-|1\rangle)$$

The Deutsch-Jozsa algorithm (cont'd)



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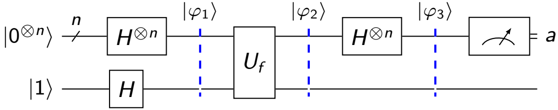
The Deutsch-Jozsa algorithm (cont'd)



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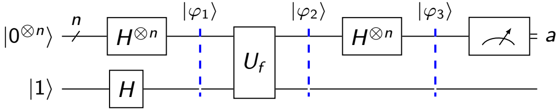
\end{aligned}$$

The Deutsch-Jozsa algorithm (cont'd)



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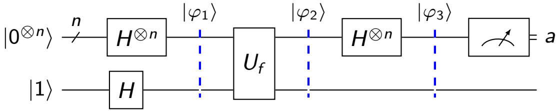
\end{aligned}
$$

The Deutsch-Jozsa algorithm (end)



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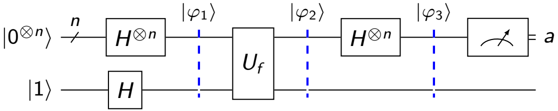
What is the probability P_0 of reading 0^n after measuring the n first qubits?

The Deutsch-Jozsa algorithm (end)



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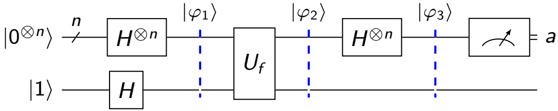

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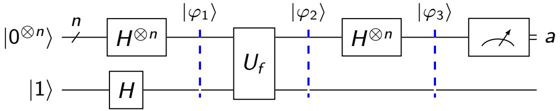
The Deutsch-Jozsa algorithm (end)



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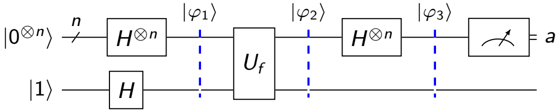
- If f is balanced then necessarily $P_0 = 0$

The Deutsch-Jozsa algorithm (end)



$$\varphi_3 = \left(\sum_{\vec{y} \in \mathbb{Z}_2^n} \left(\frac{1}{2^n} \sum_{\vec{x} \in \mathbb{Z}_2^n} (-1)^{f(\vec{x})} (-1)^{\vec{x} \odot \vec{y}} \right) |\vec{y}\rangle \right) \otimes |-\rangle$$

What is the probability P_0 of reading 0^n after measuring the n first qubits?

- We have

$$P_0 = \left(\frac{1}{2^n} \sum_{\vec{x} \in \mathbb{Z}_2^n} (-1)^{f(\vec{x})} \right)^2$$

- If f is constant then necessarily $P_0 = 1$
- If f is balanced then necessarily $P_0 = 0$

Conclusion

The single query to U_f permits to determine whether f is constant or balanced